Flight Phase Identification and Other Topics Related to Anomaly Detection of Flight Data

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Abstract

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This paper serves as a reference, documenting flight phase identification and other topics related to anomaly detection of flight data. The flight phase identification is needed to preprocess the National Aeronautics and Space Administration (NASA) DASHlink Sample Flight data set. Primarily, the phase as provided in the data set from the Aircraft Condition Management System (ACMS) has some noise which misclassifies flight phase. A Finite State Model (FSM) was developed to condition the ACMS signal to provide eight distinct flight phases: preflight, taxi, takeoff, climb, cruise, approach, and rollout. The algorithm is intended to be included in the data preprocessing step prior to using the data set for machine learning studies. Additionally, this paper provides a reference for linear first-order approximations from first-principles for each phase of flight.

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List of Acronyms and Acronyms

AI/ML	Artificial Intelligence and Machine Learning
AUROC	Area Under the Receiver Operator Characteristic curve
FDR	Flight Data Recorder
PINN	Physics-informed Neural Network
PR-AUC	Precision-Recall Curve
DHS	Department of Homeland Security
DoD	Department of Defense
SA	Situational Awareness
TSAD	Time-Series Anomaly Detection
NASA	National Aeronautics and Space Administration
FAA	Federal Aviation Administration
IATA	International Air Transport Association
ICAO	International Civil Aviation Organization
U.S.	United States

Glossary of Key Terms

Artificial Neural Network	Distributed computation model that abstracts the data presentation and processing of a human brain as an interconnected set of layers of network nodes called neurons
Flight Data	Information about an aircraft flight that is typically recorded by a flight data recorder. It contains data such as air speed, altitude, heading, engine parameters, control inputs, etc.
Situational Awareness (SA)	Situational awareness is the accurate perception and understanding of all the factors and conditions within the five fundamental risk elements (flight, pilot, aircraft, environment, and type of operation that comprise any given aviation situation) that affect safety before, during, and after the flight (Federal Aviation Association [FAA], 2023).
Physics-informed Learning	Refers to adding constraints to the model learning process based on physical and/or engineering principles
Phases of flight/Flight phases	Referring to the various operational states of aircraft during a flight. These are typically taxi, takeoff, en-route, approach, and landing
Runway Incursion	Presence of unexpected aircraft, vehicle or person in the protected area used for landing and takeoff

1.0 Introduction

1.1 Background

Since the flight dynamics of the aircraft vary significantly depending on the flight phase, the identification is a critical first step of preprocessing before analyzing the flight data.

2.0 Flight Phases

The Federal Aviation Administration (FAA) and International Civil Aviation Organization (ICAO) established the Commercial Aviation Safety Team / International Civil Aviation Organization Common Taxonomy Team (CICTT) to standardize the flight phase definitions (CICTT 2013; ICAO 2013). The CICTT standard describes twelve phases of flight. Additionally, it defines five subphases for the enroute phase as shown in

Table 1. Phases of Flight Definitions

CICTT Phase	Definition	Subphase	Praxis
Standing (STD)	Prior to pushback or taxi, or after arrival, at the gate, ramp, or parking area, while the aircraft is stationary.		Preflight
Pushback/Towing (PBT)	Aircraft is moving in the gate, ramp, or parking area, assisted by a tow vehicle (tug).		
Taxi (TXI)	The aircraft is moving on the aerodrome surface under its own power prior to takeoff or after landing.		Taxi
Takeoff (TOF)	From the application of takeoff power, through rotation and to an altitude of 35 feet above runway elevation.		Takeoff
Initial Climb (ICL)	From the end of the Takeoff subphase to the first prescribed power reduction, or until reaching 1,000 feet above runway elevation or the VFR pattern, whichever comes first.		Climb
En Route (ENR)	• Instrument Flight Rules (IFR): From completion of Initial Climb through cruise altitude and completion of controlled descent to the Initial Approach Fix (IAF). • Visual Flight Rules (VFR): From completion of Initial Climb through cruise and controlled descent to the VFR pattern altitude or 1,000 feet above runway elevation, whichever comes first.	Climb	
		Cruise	Cruise
		Change of Cruise Level	
		Descent	Approach
		Holding	
Maneuvering (MNV)	Low altitude/aerobatic flight operations.		

Approach (APR)	Instrument Flight Rules (IFR): From the Initial Approach Fix (IAF) to the beginning of the landing flare. Visual Flight Rules (VFR): From the point of VFR pattern entry, or 1,000 feet above the runway elevation, to the beginning of the landing flare.		Approach
Landing (LDG)	From the beginning of the landing flare until aircraft exits the landing runway, comes to a stop on the runway, or when power is applied for takeoff in the case of a touch-and-go landing.		Rollout
Emergency Descent (EMG)	A controlled descent during any airborne phase in response to a perceived emergency situation.		
Uncontrolled Descent (UND)	A descent during any airborne phase in which the aircraft does not sustain controlled flight.		
Post-Impact (PIM)	Any of that portion of the flight which occurs after impact with a person, object, obstacle or terrain.		
Unknown (UNK)	Phase of flight is not discernible from the information available.		Unknown

For this study, several of the CICTT phases have been combined into a single phase. For example, the standing and pushback/towing phases have been combined into a preflight phase. Similarly, this study combines the initial climb and climb phases into a single climb phase. Additionally, the CICTT cruise subphases have been combined into the single cruise phase. It should be noted that the change of cruise level will be referred to as step-up and step-down climbs in this praxis. The CICTT descent and holding subphases and CICTT approach phase have been combined into a single approach phase. The CICTT landing phase is closely mapped to the rollout phase. Rather than trying to detect the landing flare, this study defines the rollout phase from the touchdown point to the subsequent taxi phase. Consequently, the approach phase will conclude with the landing flare prior to rollout. The remaining phase categories of maneuvering,

emergency descent, uncontrolled descent, post-impact if they exist in the data should be detected as flight anomalies.

For time-series analysis, the dataset needs to be segmented into flights to establish temporal sequencing. Because commercial airlines typically follow a routine flight pattern following well-defined phases, the flight data must be broken into phases. Specifically for anomaly detection for commercial airlines, the flight data are pre-segmented based on the established flight phase definitions. This aligns with the fact that for commercial (fixed-wing) aircraft operations change dramatically through the flight but there is a strong similarity across flights for the same phase (Chin et al., 2019; Gavrilovski et al., 2016).

To promote phase segmentation, the NASA DASHlink Sample Flight dataset has a flight phase signal PH 1 hertz signal that is provided by the on-board Aircraft Condition Monitoring System (Matthews, 2012). There are eight phases represented as unknown, preflight, taxi, takeoff, climb, cruise, approach, rollout. The sensor signal had some intermittent noise that misclassified the phase and could potentially cause temporal sequencing issues. One approach is to develop a Finite State Machine (FSM) to determine the aircraft flight phases and transitions directly from aircraft parameters (Soler et al., 2011). This reference followed this approach and developed FSM to resolve database limitation L1. Figure 1. Architecture For Computing Stabilized Flight Phases shows the states defined along with transition criteria.

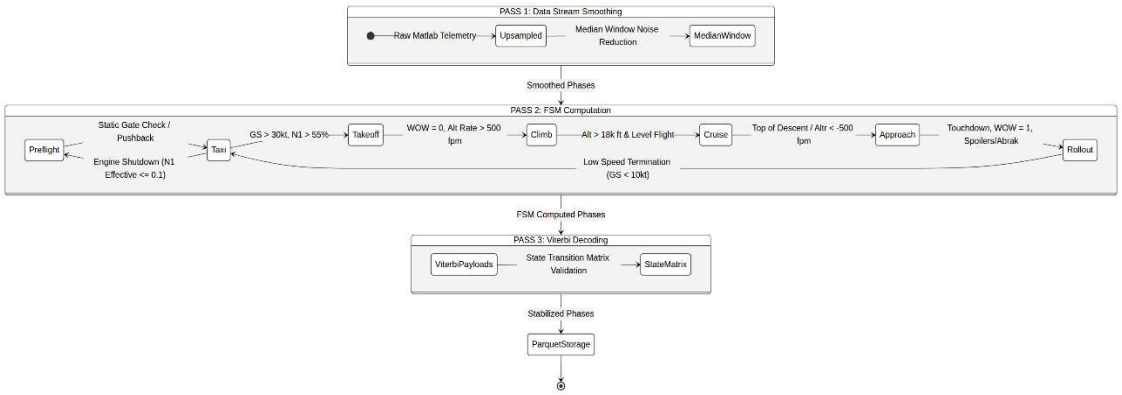


Figure 1. Architecture For Computing Stabilized Flight Phases

The FSM algorithm evolved into a two-pass system. The first pass up-sampled the input sensor signals using a zero-order hold to 16 Hz and computed a median value over 3 second windows to smooth any sensor noise. The second pass computed the flight phase confidence at each time step based on aircraft state. Additionally, some hysteresis and dwell thresholds were applied to help reduce any noise in the computed phase signal. For nominal flights the flight phase computed by the FSM was more stable than the Aircraft Condition Monitoring System (ACMS) phase. However, for various conditions like climbing or descending to a new cruise altitude, the FSM logic required further refinement.

. Fuzzy logic can be applied to the continuous flights identified by the DBSCAN algorithm to determine the flight phases (Arts, 2021; Sun et al., 2017). The algorithm was codified into an open source Python library called OpenAP (Sun et al., 2020).

The flight phase identification library of OpenAP determines the flight phase using time, altitude, speed and vertical rate inputs to the fuzzy logic algorithm. This does not utilize the other sensors provided by the dataset like weight-on-wheels, landing gear status, and flap positions. Another issue with fuzzy logic algorithms is that they are

sensitive to the noise of flight data and result in many transitions (Perrichon et al., 2024). Another limitation of the OpenAP algorithm is it focuses on in-flight operations and only has a single ground phase that does not distinguish the ground phases of preflight, taxi, and rollout. Additionally, OpenAP defines a leveling phase which for this study is considered a subphase of climb, cruise, or descent. Because of these limitations, This algorithm incorporated into the FSM to provide a guard rail for step-up and step-down conditions that establish a new level flight altitude.

To evaluate the phase identification process, the number of phase transitions N_{Trans} and the number of invalid transitions E_{Trans} can be computed. The number of phase transitions is the sum of the phase differences between two points. The number of invalid transitions is the sum of all phase transitions that violate the state transition matrix. Evaluating the ACMS phase signal and the FSM computed phase signal using these metrics for the first mission for aircraft 652, the FSM computed phase signal reduces the invalid phase transitions from the ACMS phase signal. However, for some flights, the FSM produces invalid phase transitions.

Table 2. Three-Pass Phase Stabilization

ACMS Phase Sensor		FSM Computed Phase		Final Stabilized Phase	
N_{Trans}	E_{Trans}	N_{Trans}	E_{Trans}	N_{Trans}	E_{Trans}

To further stabilize the phase, an additional pass was implemented using Hidden Markov Model to define the state transitions and the Viterbi decoder(Perrichon et al., 2024). The three-pass smoothing algorithm was applied to the ACMS flight phase to

compute a final stabilized flight phase. The final stabilized flight phase is used as part of the dataset pre-preprocessing pipeline to segment the flight data to train each machine learning model by flight phase.

2.1 Inputs to the Stabilized Flight Phase Architecture

This section describes the inputs to the Finite State Machine (FSM) that is used to compute the stabilized flight phases using the three-pass architecture described in section **Error! Reference source not found.**. The inputs are described in the following table along with usage in the algorithm:

Sensors	Description	Status	Usage
WOW	Weight-on-Wheels (0=Ground, 1=Airborne)	Active	Ground/Airborne transition
GS	Groundspeed (knots)	Active	Taxi, rollout, and takeoff velocity thresholds
CASS	Selected computed airspeed (knots)	Active	Pilot intent and profile check
BLAC/blac_smooth	Raw longitudinal acceleration and filtered (ft/sec^2)	Active	Inertial movement detection
ALT	Barometric altitude MSL (feet)	Active	Cruise floor and max altitude tracking
ALTR	Altitude rate (ft/min)	Active	Climb and approach verification
RALT	Radar Altimeter AGL (feet)	Active	Terminal approach and landing thresholds
N1C/N1T/N1CO	Engine fan speed current (%RPM), target (%RPM) and compensation flag	Active	Effective thrust, high/low power thresholds
PLA1, PLA2, PLA3, PLA4	Power lever angles for each engine (deg)	Active	Throttle position tracking
SPLG, SPLY	Spoiler deploy green, Spoiler deploy yellow	Active	Spoiler deployment for rollout
ABRK	Airbrake position (deg)	Active	Deceleration confirmation for rollout

FLAP	Flap handle angle (deg)	Active	Takeoff confidence boosting
APFD, VMODE, LMODE	Flight director status, vertical mode, lateral mode	Active	Automation engaged or pilot-in-control
ALTS	Selected altitude MSL (feet)	Active	Pilot intent
acms_phase (PH)	Phase determined by onboard ACMS	Active	
openap_macro	OpenAP phase calculation, 1Hz (CL, CR, LVL, DE)	Active	Phase guard and noise filter
LGDN, LGUP	Landing gear down (=0), Landing gear up (=1)	Inactive	Used for plotting verification of WOW only not in FSM
IVV	Inertial vertical velocity (ft/min)	Inactive	Using ALTR in FSM instead
FADF, FADS	FADEC fail all engines, FADEC status all engines	Inactive	Initial attempt to determine engine status

The third pass uses the Viterbi encoding which defines valid state transitions to enforce valid state transitions and preventing invalid transitions like cruise to rollout without approach. The following table describes the state transitions matrix:

From/To	0: Unknown	1: Preflight	2: Taxi	3: Takeoff	4: Climb	5: Cruise	6: Approach	7: Rollout
0: Unknown	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
1: Preflight	Yes	Yes	Yes	No	No	No	No	No
2: Taxi	Yes	No	Yes	Yes	No	No	No	No
3: Takeoff	Yes	No	No	Yes	Yes	No	No	No
4: Climb	Yes	No	No	No	Yes	Yes	No	No
5: Cruise	Yes	No	No	No	No	Yes	Yes	No
6: Approach	Yes	No	No	No	No	No	Yes	Yes
7: Rollout	Yes	No	Yes	No	No	No	No	Yes

3.0 Other Topics

3.1 Rigid-Body Equations and Linearization

This section provides more detailed information about the derivation of the rigid-body equations of motion from first principles and the simplifications involved to highlight the important features by flight phase (Nelson, 1998).

3.2 Non-Linear Equations of Motion in Inertial Frame

The following equations apply Newton's second law to model an aircraft translational acceleration and rotational acceleration:

$$\begin{aligned} \vec{F} &= \frac{d}{dt}\left(m\vec{V}\right) \\ \vec{M} &= \frac{d}{dt}\left(\vec{H}\right) \end{aligned} \tag{1}$$

where F is the external forces acting on the aircraft, m is the mass of the aircraft, V is the velocity of the aircraft, M is the total moments acting on the aircraft and H is the total angular momentum of the aircraft. Assuming constant mass m, constant aircraft inertia tensor I and moving body-axis reference frame,

$$\begin{aligned} \vec{F} &= m\left(\left.\frac{d\vec{V}}{dt}\right|_B + \vec{\Omega} \times \vec{V}\right) \\ \vec{M} &= \left.\frac{d\left(\vec{I} \cdot \vec{\Omega}\right)}{dt}\right|_B + \vec{\Omega} \times \left(\vec{I} \cdot \vec{\Omega}\right) \end{aligned} \tag{2}$$

where $\vec{V} = [u, v, w]^T$ is the total velocity vector in the body-axis frame, $\vec{\Omega} = [p, q, r]^T$ is the total angular velocity vector in the body-axis frame, $\vec{F} = [F_X, F_Y, F_Z]^T$ is the total external aerodynamic, thrust and gravity forces acting on the aircraft, $\vec{M} = [L, M, N]^T$ is the external aerodynamic, thrust, and gravity moments acting on the aircraft, $\vec{H} = \vec{I} \cdot \vec{\Omega}$ is

the angular momentum vector and I is the rigid-body inertia tensor. If the xz-plane is assumed to be the plane of symmetry of the aircraft, the products of inertia become $I_{yz} = I_{xy} = 0$. This simplifies the moment equations. The scalar force and moment equations are:

$$\begin{aligned} \mathrm{F}_X - mg\sin\theta &= m(\dot{u} + qw - rv) \\ \mathrm{F}_Y + mg\sin\phi\cos\theta &= m(\dot{v} + ru - pw) \\ \mathrm{F}_Z + mg\cos\phi\cos\theta &= m(\dot{w} + pv - qu) \\ L &= I_{xx}\dot{p} - I_{xz}\dot{r} + qr(I_{zz} - I_{yy}) - I_{xz}pq \\ M &= I_{yy}\dot{q} + rq(I_{xx} - I_{zz}) + I_{xz}(p^2 - r^2) \\ N &= -I_{xz}\dot{p} + I_{zz}\dot{r} + pq(I_{yy} - I_{xx}) + I_{xz}qr \end{aligned} \tag{3}$$

The position and orientation of the aircraft must be related to a fixed frame of reference such that at $t = 0$, the fixed frame and the moving frame coincide. Three consecutive rotations can describe the attitude of the aircraft. The choice of the direction cosine matrix is arbitrary but for aircraft the heading-pitch-roll (3-2-1) is typical. Consequently, the first rotation of heading (ψ) is defined about the z-axis, the second rotation of pitch (θ) about the y-axis, and the third rotation of roll (ϕ) is about the x-axis, the fixed-frame velocity can be related back to the Euler angles and velocity components:

$$\frac{d\vec{R}}{dt} = \vec{R}_{i/b}\vec{V}_B = \vec{R}_1(\phi)\vec{R}_2(\theta)\overrightarrow{R_3}(\psi)\vec{V}_B \tag{4}$$

$$\begin{bmatrix} \frac{dx}{dt} \\ \frac{dy}{dt} \\ \frac{dz}{dt} \end{bmatrix} = \begin{bmatrix} \cos\theta\cos\psi & \sin\phi\sin\theta\cos\psi - \cos\phi\sin\psi & \cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi \\ \cos\theta\sin\psi & \sin\phi\sin\theta\sin\psi + \cos\phi\cos\psi & \cos\phi\sin\theta\sin\psi - \end{bmatrix} \begin{bmatrix} u \\ v \\ w \end{bmatrix} \tag{5}$$

where $\vec{R} = [x, y, z]^T$ is the position vector in the fixed frame and $\vec{V}_B = [u, v, w]^T$ is the velocity vector in the body-axis frame. By integrating this equation with respect to time, the airplane's position in the fixed frame is determined.

Similarly, for the Euler angle vector E in the fixed inertial frame, the direction cosine matrix $\vec{R}_{b/i}$ is the sum of the angular velocities in each direction in the body-axis.

$$\vec{\Omega} = \vec{R}_{b/i}\frac{d\vec{E}}{dt} = \left(\begin{bmatrix}\dot{\phi}\\0\\0\end{bmatrix} + \begin{bmatrix}1 & 0 & 0\\0 & \cos\phi & \sin\phi\\0 & -\sin\phi & \cos\phi\end{bmatrix}\begin{bmatrix}0\\\dot{\theta}\\0\end{bmatrix} + \begin{bmatrix}1 & 0 & 0\\0 & \cos\phi & \sin\phi\\0 & -\sin\phi & \cos\phi\end{bmatrix}\begin{bmatrix}\cos\theta & 0 & -\sin\theta\\0 & 1 & 0\\\sin\theta & 0 & \cos\theta\end{bmatrix}\begin{bmatrix}0\\0\\\dot{\psi}\end{bmatrix}\right)\frac{d\vec{E}}{dt} \quad (6)$$

$$\begin{bmatrix}p\\q\\r\end{bmatrix} = \begin{bmatrix}1 & 0 & -\sin\theta\\0 & \cos\phi & \sin\phi\cos\theta\\0 & -\sin\phi & \cos\phi\cos\theta\end{bmatrix}\begin{bmatrix}\dot{\phi}\\\dot{\theta}\\\dot{\psi}\end{bmatrix} \quad (7)$$

where $\vec{\Omega} = [p \quad q \quad r]^T$ is the angular velocity vector in body-axis and $\frac{d\vec{E}}{dt} = [\dot{\phi} \quad \dot{\theta} \quad \dot{\psi}]^T$ is the Euler rates in the fixed inertial axis. Inverting the direction cosine matrix and solving for the Euler rates:

$$\begin{bmatrix}\dot{\phi}\\\dot{\theta}\\\dot{\psi}\end{bmatrix} = \begin{bmatrix}1 & \sin\phi\tan\theta & \cos\phi\tan\theta\\0 & \cos\phi & -\sin\phi\\0 & \sin\phi\sec\theta & \cos\phi\sec\theta\end{bmatrix}\begin{bmatrix}p\\q\\r\end{bmatrix} \quad (8)$$

Integrating this expression will compute the Euler rates.

The equations can be linearized using small-disturbance theory where each variable is replaced by a reference value plus a perturbation.

$$\begin{aligned}\vec{V} &= \vec{V}_0 + \Delta\vec{V}\\\vec{\Omega} &= \vec{\Omega}_0 + \Delta\vec{\Omega}\\\vec{F} &= \vec{F}_0 + \Delta\vec{F}\\\vec{\delta} &= \vec{\delta}_0 + \Delta\vec{\delta}\end{aligned} \quad (9)$$

where the velocity perturbation vector is $\Delta\vec{V} = [\Delta u, \Delta v, \Delta w]^T$, the reference velocity vector is $\overrightarrow{V_0} = [U_0, V_0, W_0]^T$, the moment perturbation vector is $\Delta\vec{\Omega} = [\Delta L, \Delta M, \Delta N]^T$, the reference momentum vector is $\overrightarrow{\Omega_0} = [L_0, M_0, N_0]^T$, the force perturbation vector is $\Delta\vec{\mathrm{F}} = [\Delta F_X, \Delta F_Y, \Delta F_Z]^T$, the reference momentum vector is $\overrightarrow{\mathrm{F_0}} = [F_{X0}, F_{Y0}, F_{Z0}]^T$.

For example, the longitudinal translational and rotational equations, this becomes

$$(\mathrm{F_{X0}} + \Delta\mathrm{F_X}) - \mathrm{mg}\sin(\theta_0 + \Delta\theta) = \mathrm{m}\left(\frac{\mathrm{d}(\mathrm{u_0} + \Delta\mathrm{u})}{\mathrm{dt}} + (\mathrm{q_0} + \Delta\mathrm{q})(\mathrm{w_0} + \Delta\mathrm{w}) - (\mathrm{r_0} + \Delta\mathrm{r})(\mathrm{v_0} + \Delta\mathrm{v})\right) \quad (10)$$

$$(L_0 + \Delta L) = I_{xx}\frac{d(p_0 + \Delta p)}{dt} - I_{xz}\frac{d(r_0 + \Delta r)}{dt} + (q_0 + \Delta q)(r_0 + \Delta r)\left(I_{zz} - I_{yy}\right) - I_{xz}(p_0 + \Delta p)(q_0 + \Delta q) \quad (11)$$

Assuming steady, symmetric, straight and level reference trim condition, all time derivatives will be zero ($\frac{du_o}{dt}, \frac{dv_o}{dt}, \cdots = 0$) and angular velocities are zero ($p_0, q_o, r_o = 0$). The trim values for lateral velocity (v_0) and roll angle (ϕ_0) are zero. Aligning the body-axis with longitudinal axis at trim sets the vertical velocity (w_0) to zero. Furthermore, assuming small angle approximations with sine and cosine sum identities:

$$\sin(\theta_0 + \Delta\theta) = \sin\theta_0\cos\Delta\theta + \cos\theta_0\sin\Delta\theta \approx \sin\theta_0 + \cos\theta_0 \cdot \Delta\theta$$

$$\cos(\theta_0 + \Delta\theta) = \cos\theta_0\cos\Delta\theta - \sin\theta_0\sin\Delta\theta \approx \cos\theta_0 - \sin\theta_0 \cdot \Delta\theta$$

On the left-hand side of the equations, the force and moment changes in the longitudinal directions can be expanded using the Taylor series expansion, keeping only the linear terms and assuming that the longitudinal forces are only a function of u, w, elevator, and thrust and longitudinal moment a function of *v, p, r*, rudder and ailerons then

$$\Delta F_X = \frac{\partial X}{\partial u}\Delta u + \frac{\partial X}{\partial w}\Delta w + \frac{\partial X}{\partial \delta_e}\Delta\delta_e + \frac{\partial X}{\partial T}\Delta\delta_T \quad (12)$$

$$\Delta L = \frac{\partial L}{\partial v}\Delta v + \frac{\partial L}{\partial p}\Delta p + \frac{\partial L}{\partial r}\Delta r + \frac{\partial L}{\partial \delta_r}\Delta\delta_r + \frac{\partial L}{\partial \delta_a}\Delta\delta_a \quad (13)$$

Similarly, this Taylor series expansion can be done for the other force and moment equations. After substituting the force and moment changes equations back into the small-disturbance equations, it turns out that the longitudinal and lateral equations can be decoupled into longitudinal equations and lateral-directional equations. For convenience, these are shown in matrix form:

$$\begin{bmatrix}\Delta\dot{u}\\ \Delta\dot{w}\\ \Delta\dot{q}\\ \Delta\dot{\theta}\end{bmatrix} = \begin{bmatrix} X_u & X_w & 0 & -g\cos\theta_0 \\ Z_u & Z_w & U_0 + Z_q & -g\sin\theta_0 \\ M_u + M_{\dot{u}}X_u & M_w + M_{\dot{u}}Z_w & M_q + M_{\dot{u}}(U_0 + Z_q) & -M_{\dot{u}}g\sin\theta_0 \\ 0 & 0 & 1 & 0 \end{bmatrix}\begin{bmatrix}\Delta u\\ \Delta w\\ \Delta q\\ \Delta\theta\end{bmatrix} + \begin{bmatrix} X_{\delta_e} & X_{\delta_b} & X_{\delta_T} \\ Z_{\delta_e} & Z_{\delta_b} & Z_{\delta_T} \\ M_{\delta_e} + M_{\dot{u}}X_e & M_{\delta_b} + M_{\dot{u}}X_b & M_{\delta_T} + M_{\dot{u}}X_T \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix}\Delta\delta_e\\ \Delta\delta_b\\ \Delta\delta_T\end{bmatrix} \quad (14)$$

$$\begin{bmatrix}\Delta\dot{v}\\ \Delta\dot{p}\\ \Delta\dot{r}\\ \Delta\dot{\phi}\\ \Delta\dot{\psi}\end{bmatrix} = \begin{bmatrix} Y_v & Y_p & Y_r - U_0 & g\cos\theta_0 & 0 \\ L_v^* & L_p^* & L_r^* & 0 & 0 \\ N_v^* & N_p^* & N_r^* & 0 & 0 \\ 1 & \tan\theta_0 & 0 & 0 & 0 \\ 0 & 0 & \sec\Theta_0 & 0 & 0 \end{bmatrix}\begin{bmatrix}\Delta v\\ \Delta p\\ \Delta r\\ \Delta\phi\\ \Delta\psi\end{bmatrix} + \begin{bmatrix} Y_{\delta_a} & Y_{\delta_r} & Y_{\delta_{steer}} \\ L_{\delta_a}^* & L_{\delta_r}^* & L_{\delta_{steer}}^* \\ N_{\delta_a}^* & N_{\delta_r}^* & N_{\delta_{steer}}^* \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix}\Delta\delta_a\\ \Delta\delta_r\\ \Delta\delta_{steer}\end{bmatrix} \quad (15)$$

where Δu, Δv, Δw is the perturbations in forward speed, vertical speed, and vertical speed; $\Delta\dot{u}$, $\Delta\dot{v}$, $\Delta\dot{w}$ are the perturbation velocity rates; θ_o is the reference pitch angle; g is the acceleration due to gravity; the stability derivatives normalized by mass are $X_u = \frac{1}{m}\frac{\partial X}{\partial u}$, $X_w = \frac{1}{m}\frac{\partial X}{\partial w}$, $Z_u = \frac{1}{m}\frac{\partial Z}{\partial u}$, $Z_w = \frac{1}{m}\frac{\partial Z}{\partial w}$, $Z_q = \frac{1}{m}\frac{\partial Z}{\partial q}$; the stability derivatives normalized by mass with respect to the elevator, brakes, and thrust are $X_{\delta_e} = \frac{1}{m}\frac{\partial X}{\partial \delta_e}$, $X_{\delta_b} = \frac{1}{m}\frac{\partial X}{\partial \delta_b}$, $X_{\delta_T} = \frac{1}{m}\frac{\partial X}{\partial \delta_T}$, $Z_{\delta_e} = \frac{1}{m}\frac{\partial Z}{\partial \delta_e}$, $Z_{\delta_b} = \frac{1}{m}\frac{\partial Z}{\partial \delta_b}$, $Z_{\delta_T} = \frac{1}{m}\frac{\partial Z}{\partial \delta_T}$, the stability derivatives that are normalized by inertia are $M_u = \frac{1}{I_y}\frac{\partial M}{\partial u}$, $M_{\dot{u}} = \frac{1}{I_y}\frac{\partial M}{\partial \dot{u}}$, $M_q = \frac{1}{I_y}\frac{\partial M}{\partial q}$; the stability derivatives normalized by inertia with respect to the elevator, brakes and thrust are $M_{\delta_e} = \frac{1}{I_y}\frac{\partial M}{\partial \delta_e}$, $M_{\delta_b} = \frac{1}{I_y}\frac{\partial M}{\partial \delta_b}$, $M_{\delta_T} = \frac{1}{I_y}\frac{\partial M}{\partial \delta_T}$; $\Delta p, \Delta q, \Delta r$ are the

perturbation angular rates; $\Delta\dot{p}, \Delta\dot{q}, \Delta\dot{r}$ are perturbation angular accelerations; $L^* = L + \frac{I_{xz}}{I_x} N$ and $N^* = N + \frac{I_{xz}}{I_z} L$ are the cross-coupling derivatives; Y, L, N are the side-force, rolling and yawing moment derivatives; $\Delta\delta_a, \Delta\delta_r, \Delta\delta_{steer}$ are the aileron, rudder and steering inputs.

3.3 Estimating Business Jet Moments of Inertia Using Roskam's Class I Method

The moments of inertia using Roskam's Class I method is used for preliminary aircraft design. The following provides the details for estimating the moments of inertia from Roskam's Class I method for a business jet class aircraft (Pegram & Anemaat, 2000).

1. Aircraft Data
    a. Mass (m): 42, 184 kg maximum takeoff weight
    b. Wingspan (b): 26.34 m
    c. Fuselage length (L): 28.55 m
2. Identify Empirical Coefficients
    a. Roll factor ($\bar{R}_x$): 0.25 for wing-mounted jet engines
    b. Pitch factor ($\bar{R}_y$): 0.38 for typical jet transport
    c. Yaw factor ($\bar{R}_z$): 0.39 for transport category structures
3. Compute Roll Inertia

$$I_{xx} = m \cdot (\bar{R}_x \cdot b)^2 = 42184 \cdot (0.25 \cdot 26.34)^2 = 1829192.1\ kg \cdot m^2$$

4. Compute Pitch Inertia

$$I_{yy} = m \cdot \left(\bar{R}_y \cdot L\right)^2 = 42184 \cdot (0.38 \cdot 28.55)^2 = 4965090.6\ kg \cdot m^2$$

5. Compute Yaw Inertia

$$I_{zz} = m \cdot \left(\bar{R}_z \cdot \frac{b+L}{2}\right)^2 = 42184 \cdot (0.39 \cdot \frac{26.34 + 28.55}{2})^2$$

$$= 4832851.4\ kg \cdot m^2$$

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